

Old tree root channels in acid soils in the humid tropics: Important for crop root penetration, water infiltration and nitrogen management

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Abstract

Under high rainfall conditions on acid soils with shallow crop root systems the rate of N leaching is high. A simple model predicts nitrogen uptake efficiency as a function of the amount of rainfall in excess of evapotranspiration, rooting depth and degree to which N leaching is retarded in comparison with water transport. Field observations on acid soils in S.E. Nigeria and S. Sumatera (Indonesia) showed that this model should be amended to include the role of old tree root channels. Crop roots can follow these channels, which are coated with partly decayed organic matter, into the acid subsoil. Measurements of water infiltration with a Guelph permeameter and a methylene blue dye showed that such channels form the major infiltration sites during rainstorms. Implications for nitrogen use efficiency and cropping pattern are discussed.

Introduction

In the humid tropics acid soils dominate. Efficient utilization of nitrogen by natural vegetation and/or crops under these conditions is hindered by high rates of leaching and by shallow rooting due to subsoil toxicity. The resulting low nitrogen supply is often a major constraint to plant production. Use of fertilizer N is limited by a low agronomic efficiency, unfavourable cost/benefit ratios, poor accessibility and increasing concern for environmental effects. To obtain a higher agronomic N use efficiency, basic soil and plant processes should be known.

Van Noordwijk (1989) presented a simple model for N leaching and uptake by crops under conditions of high rainfall. The model is based on one-dimensional flow of water, nitrogen and other nutrients in the profile, driven by the excess of rainfall over evapotranspiration. Recovery of nutrients is possible within an 'effective rooting depth', defined for each nutrient on the basis of a root length density (L_{rv} , cm cm^{-3})

nutrients at the rate required by the crop (De Willigen and Van Noordwijk, 1987). Various patterns of nutrient release in the topsoil during the growing season were compared to study the relative importance of *synchronization* of nutrient release and crop demand. As shown in Table 1, the rooting depth required to make a nutrient recovery of at least 80% possible, depends on the excess rainfall and the retardation factor, K_a .

Retardation of nutrient movement with respect to water flow can be based on three types of mechanism:

a) Interactions between solid phase and soil solution (adsorption or ion exchange). Adsorption constants are generally much higher for ammonium than for nitrate, so the rate of nitrification is important for potential N leaching. In a model incorporating pH-dependent nitrification, De Willigen (1985) showed that increasing the pH of acid soils may decrease N use efficiency by crops.

b) Microbiological immobilization/mineraliza-

Table 1. Model calculations of the influence of the annual surplus of infiltration over evapotranspiration, $L_w(1)$ [m], on the leaching depth after 1 year $Z_l(1)$ [m] and on the effective rooting depth Z_r [m] required for 80% interception of a nutrient. Calculations are presented for three values of the apparent adsorption constant K_a [mL cm^{-3}] (roughly corresponding to nitrate, ammonium and potassium, respectively) and, given a steady rate of water infiltration, for two patterns of nutrient release in the available form: relatively early and relatively late in the growing season (Van Noordwijk, 1989)

$L_w(1)$	$K_a = 0$			$K_a = 3$			$K_a = 10$		
	$Z_l(1)$	Z_r early	Z_r late	$Z_l(1)$	Z_r early	Z_r late	$Z_l(1)$	Z_r early	Z_r late
0.25	0.75	0.67	0.41	0.19	0.17	0.10	0.07	0.06	0.04
0.50	1.50	1.34	0.82	0.37	0.33	0.21	0.14	0.12	0.08
0.75	2.25	2.00	1.24	0.56	0.50	0.31	0.20	0.18	0.11
1.00	3.00	2.67	1.65	0.75	0.67	0.42	0.28	0.25	0.15
1.50	4.50	4.00	2.47	1.12	1.00	0.62	0.42	0.37	0.23
2.00	6.00	5.34	3.30	1.50	1.33	0.83	0.55	0.49	0.30

tion, as studied in the TSBF programme (Anderson and Ingram, 1989).

c) Bypass flow. Most models of nutrient and water transport in soils assume that water infiltrates homogeneously into the soil. Especially in cracking soils considerable deviations from this pattern occur, which may have important consequences for leaching of nutrients. If the major part of the rain infiltrates in cracks or macropores, bypassing the bulk soil, leaching rates of nutrients can be low, provided that they are incorporated into the bulk soil (Elliott and Coleman, 1988; Grimme and Juo, 1985).

According to this conceptual model the following options are open to increase N use efficiency on acid soils in the humid tropics:

1. to increase rooting depth by selecting acid-soil-tolerant germ plasm,
2. to increase apparent adsorption of N, especially by controlling nitrification so as to keep N in the NH_4 form and by maintaining a sufficiently high CEC through a high organic matter content,
3. to speed up incorporation of fertilizer in the bulk soil and increase bypass flow of excess rainwater via macropores, or as surface runoff, the latter with the risks of soil erosion,
4. to improve synchronization of release and demand by manipulating the immobilization/mineralization balance by managing organic inputs, soil water content, soil temperature and/or the presence or absence of certain soil biota.

To apply these principles to improvement of actual cropping systems, we need background information on the present situation. In this paper emphasis will be given to rooting depth

of food crops and water infiltration in two acid tropical soils in the humid tropics. Data are presented for two sites where similar research projects on N management for food crops are conducted, one in S.E. Nigeria and one in S. Sumatera (Indonesia).

Sites and methods

Soil and climate

Observations are discussed for two sites: the high-rainfall substation of I.I.T.A. in Onne (Port Harcourt, S.E. Nigeria) and the field site in Ketapang (Lampung, S. Sumatera) of a collaborative project of Brawijaya University, Malang, Indonesia and the Institute for Soil Fertility Research, Haren, the Netherlands. The sites are comparable in annual rainfall and soil conditions. Table 2 summarizes data on soil and climate for the two sites. Figure 1 shows the predicted leaching patterns for the two sites, using the simple model of Van Noordwijk (1989).

Root observations

On both sites root observations were carried out by simply observing root distribution on profile walls or by washing monolith samples on pin-boards to evaluate root length density. The techniques used for root observation were described in Anderson and Ingram (1989).

Table 2. Climate and soil data for the two sites described, Onne in S.E. Nigeria and Ketapang in S. Sumatera, Indonesia (Hairiah and Van Noordwijk, 1989)

Depth (cm)	Onne				Ketapang		
	0–15	5–15	15–25	25–40	0–3	3–29	29–66
Soil texture							
Sand (%)	83	83	77	73	75	64	63
Silt (%)	13	13	12	11	12	13	12
Clay (%)	4	4	11	17	14	23	25
C (%)	1.78	1.31	0.98	0.73	1.73	0.43	0.18
Bulk density (g cm^{-3})	1.40	1.50	1.48	1.40		1.38	1.42
pH (KCl)					5.2	3.8	3.8
pH (H_2O)	4.5	4.5	4.5	4.5	6.1	4.6	4.6
Annual rainfall [mm]	2400				2200		

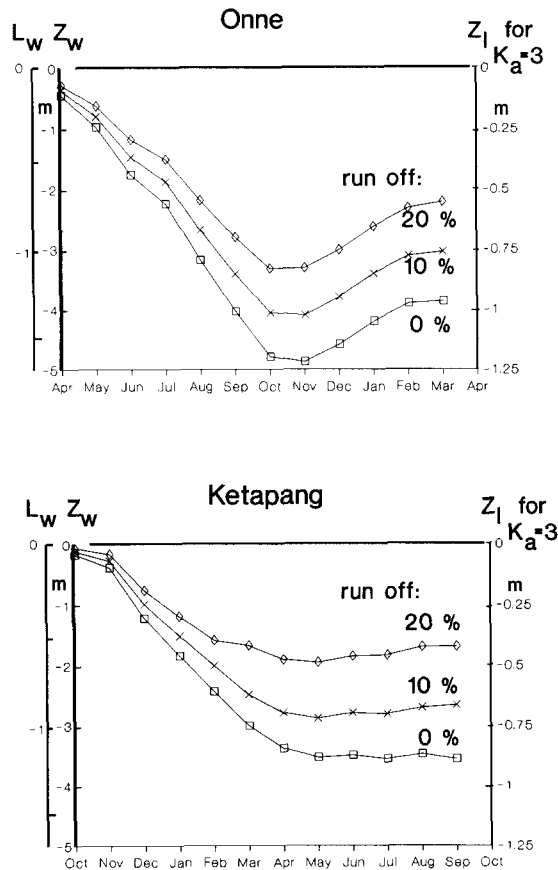


Fig. 1. Excess of rainfall over evapotranspiration, L_w , during an average growing season and depth of infiltration of water front, Z_w , for three percentages of surface runoff. The right axis represents the predicted depth of leaching fronts, Z_l , for a nutrient with a retardation factor K_a of 3 mL cm^{-3} . Climatic data for Onne (S.E. Nigeria) and Ketapang (S. Sumatera) were used for the calculations; a uniform saturated water content of 33% (v/v) was assumed.

Infiltration measurements with a Guelph permeameter

To estimate parameters for a model of the water balance, infiltration measurements were performed in Ketapang with the Guelph permeameter (Heinen and Raats, 1990). The Guelph permeameter is a Mariotte flask which can maintain a constant head in a borehole. The discharge rate is measured and the approximately steady infiltration rate after the initial transition period (of several minutes in our case) is recorded. The steady discharge, Q , can be expressed as a function of the field-saturated hydraulic conductivity, K_{fs} [$\text{L} \cdot \text{T}^{-1}$] and of the matric flux potential, ϕ_m [$\text{L}^2 \cdot \text{T}^{-1}$]. This theory is based on an infinitely large, homogeneous, isotropic and rigid flow domain at a uniform initial water content (Heinen and Raats, 1990). The two parameters can be obtained from two steady discharges out of the same borehole at two subsequent constant water levels of, say, 5 and 10 cm, yielding Q_5 and Q_{10} , respectively. Heinen and Raats (1990) showed that in order to obtain positive estimates of K_{fs} and ϕ_m , the ratio of the two steady discharges, Q_{10}/Q_5 , should fall inside a certain range. For the two models they discussed they found: $1.31 < Q_{10}/Q_5 < 2.42$ for 'model Guelph' and $1.21 < Q_{10}/Q_5 < 2.21$ for 'model Philip' (Heinen and Raats, 1990). If the actual ratio falls outside this range it indicates that the theory, based on the above-mentioned assumptions, is not applicable.

Measurements were done in secondary forest

and on plots which were cleared one or three years before the measurements. Infiltration was measured in boreholes of 20 and 50 cm deep.

Methylene blue infiltration pattern

To obtain information on the homogeneity of water infiltration, a methylene blue solution in water (0.05 g L^{-1}) was infiltrated into the soil. Infiltration was done either through metal rings of 9 cm diameter, pushed into the soil to about 1 cm, or through the standard-size bore-hole for the Guelph permeameter. The amounts infiltrated were equivalent to 100–200 mm of rain; this amount was infiltrated in about 10 minutes. Next, a small soil pit was dug adjacent to the infiltration site and three or four vertical planes in front of, through and behind the infiltration site were smoothened. Blue spots in the soil were mapped on a 1:1 scale on a transparent plastic sheet covering the soil.

Results

Root development of food crops

As shown in Table 3, the 'effective rooting depth' of food crops is small in Onne, S.E. Nigeria. Maize hardly penetrates into the subsoil beyond a depth of 15 to 20 cm; upland rice penetrates more deeply. Maize roots penetrating into the subsoil showed the typical symptoms of Al-toxicity: thick, brittle, non-elastic root tips. In the profile wall observations occasionally large channels were found, which were loosely filled with topsoil or partly decayed tree roots. In these old tree root channels crop roots of maize, upland rice and cassava were found. Remains of palm roots were common: their exodermis resists decomposition and stays behind as a tube of approximately 5 mm wide. Palm roots have a preferential horizontal orientation with a restricted rooting depth. Less frequently old root channels of dicotyledonous trees were present. One such channel was found to still have a diameter of more than 1 cm at a depth of 1.5 m. Maize roots in these old tree root channels had a normal morphology, without symptoms of Al-toxicity. To test whether the soil in the old root

Table 3. Estimates of 'effective rooting depth' Z_r for K, NH_4 and NO_3 for some food crops on an ultisol in Onne, S.E. Nigeria (based on Hairiah and Van Noordwijk, 1986); $Z_r(\text{K})$, $Z_r(\text{NH}_4)$ and $Z_r(\text{NO}_3)$ are based on root length densities of 1, 0.5 and 0.1 cm cm^{-3} , respectively

Crop	Age (weeks)	$Z_r(\text{K})$	$Z_r(\text{NH}_4)$	$Z_r(\text{NO}_3)$
		(m)		
Maize	2	0.05	0.1	0.1
	5	0.1	0.15	0.2
	8	0.1	0.2	0.25
	14	0.1	0.2	0.25
Upland rice	3	0	0.05	0.1
	5	0	0.1	0.2
	9	0.2	0.3	0.35
	14	0.2	0.3	0.77
Cowpea	5	0.1	0.1	0.1
	8	0.2	0.25	0.35
	14	0.2	0.25	0.35
Cassava	2	0*	0	0
	5	0*	0	0.2
	8	0*	0	0.3
	14	0*	0.1	0.35
	40	0*	0.2	0.6

* 30–50% of root length is in fact mycorrhiza.

channels was less acid than the surrounding subsoil, soil was collected and the pH measured. As shown in Table 4, the soil in the old root channel was at least as acid as the surrounding subsoil.

In Ketapang, S. Sumatera, similar observations were made. Maize roots were restricted to the topsoil, and the few roots penetrating beyond a depth of 20 cm again showed Al-toxicity symptoms. Occasionally maize roots happen to find a skeleton of tree roots and penetrate deeper, as shown in Figure 2. Old tree roots were observed as deep as the plinthic layer in the soil profile, at a depth of about 90 cm.

Table 4. Soil pH at three depths in Onne in the bulk soil and in loose black soil collected from old tree root channels in the layer 0.3–0.6 m depth (Hairiah and Van Noordwijk, 1986)

Depth (m)	pH (KCl)	pH (H_2O)
0–0.2	3.6	4.7
0.2–0.3	3.6	4.5
0.3–0.6	3.6	4.4
0.3–0.6R	3.4	4.2

R = collected from old tree root channel.

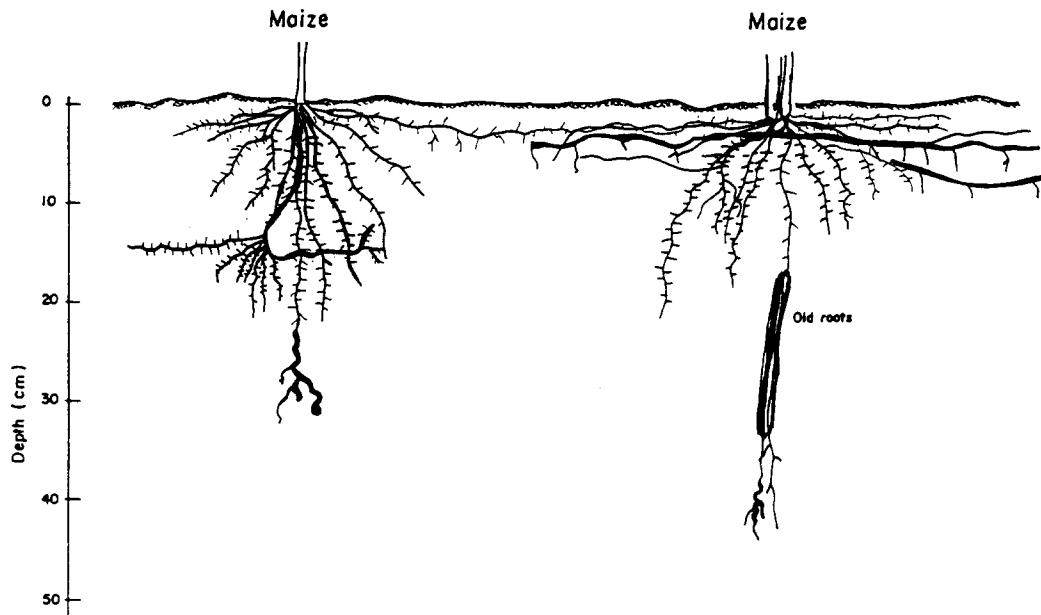


Fig. 2. Maize root distribution in Ketapang (S. Sumatera), showing thick and brittle roots in the subsoil and deeper root penetration through an old tree root skeleton.

Water infiltration

Results of a series of measurements with the Guelph permeameter in Ketapang proved to be difficult to interpret. Mostly the ratio Q_{10}/Q_5 for individual measurements was outside the range which yields positive estimates of both the saturated hydraulic conductivity K_{fs} and the matrix potential ϕ_m . Table 5 shows the average steady discharge rates of water, Q_5 and Q_{10} , into boreholes of 20 and 50 cm deep. For each of the five situations studied, the ratio of average Q_{10} and Q_5 is outside the theoretical range.

Despite a large variation, the data showed that the infiltration rate was lower at a depth of 50 cm than at a depth of 20 cm. Infiltration rates were

higher in the area under forest cover than in the plots recently cleared. Further analysis of these differences is not possible, however, because the data show that the soil physical model based on homogeneous soils is not applicable here.

Infiltration of methylene blue

On the light-colored soil of Ketapang blue staining was easily visible. Charcoaled root remains gave some problems as they were partly stained and the distinction between black and blue-black was not easy. Figure 3A shows an example of the results. Apart from an intensively colored upper 2 cm, blue staining occurred in minor cracks and old root channels, either empty or containing

Table 5. Average 'steady' discharge rates [$\text{cm}^3 \text{s}^{-1}$] into standard-sized boreholes at two depths for the Guelph permeameter, with 5 and 10 cm water level in the borehole (Q_5 and Q_{10} , respectively); n = number of observations

Location	Depth							
	20 cm				50 cm			
	Q_5	Q_{10}	Q_{10}/Q_5	n	Q_5	Q_{10}	Q_{10}/Q_5	n
Forest	0.479	1.37	2.86	10				
1-year open	0.071	0.443	6.24	10	0.0175	0.096	5.49	4
3-year open	0.088	0.393	4.46	10	0.0043	0.0131	3.03	4

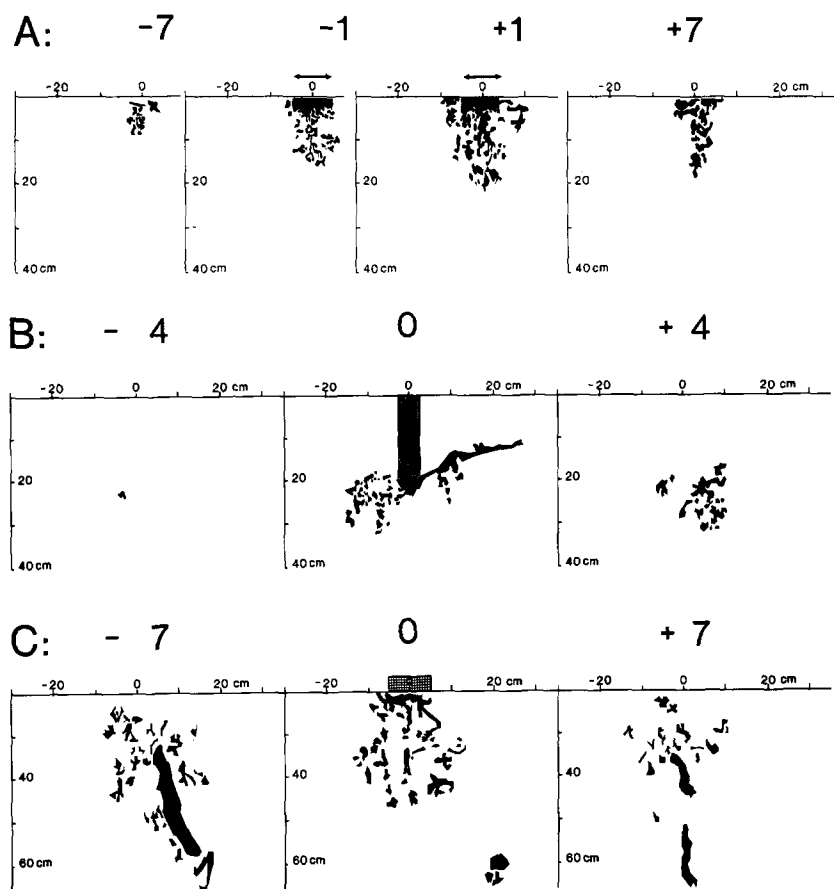


Fig. 3. Patterns of dye infiltration into the soil. Maps were made in front of, through and behind the infiltration site; A, B and C represent three cases with increasing deviation from the homogeneous soil conditions for which the Guelph parameter was developed; for further discussion see text.

charcoaled root skeletons. In some cases a diffusive blue staining occurred in cracks originating from a blue-stained root channel. On the map no blue staining occurred beyond 30 cm depth; by digging up the soil under the infiltration site one case of blue staining in a crack to a depth of 35 cm was found.

In the second set of staining observations the Guelph permeameter was used to infiltrate the dye. Figure 3B shows some results. In this case Q_{10} was 5 times as high as Q_5 . The dye pattern showed that an old tree root channel was present close to the borehole and became water-filled during the measurement of Q_{10} . In another case infiltration through a ring was found to be so slow that the measurement was stopped in one place, while a second measured at 10 cm distance

showed rapid infiltration. The dye pattern (Fig. 3C) revealed that the second measurement was performed close to an old tree stump. Coloured old root channels were found radiating from this stump into the subsoil.

Discussion

Rooting depth of crops

The observations presented here for the two sites show that old tree root channels are important for both root penetration and infiltration of water under water-saturated conditions. A comparison of Table 3 and Figure 1 shows that the simple leaching model predicts that crop rooting

depth cannot (maize) or only just (upland rice) keep up with leaching rates for nutrients with a retardation factor K_a of 3 mL cm^{-3} . Any increase in rooting depth or in retardation factor, *e.g.* by increased bypass flow, can therefore increase N recovery by the crop.

One may wonder why crop roots can follow these channels without symptoms of Al-toxicity while the pH of the loose soil in the channels is even lower than that in the surrounding subsoil. It might be that the coating of organic matter around the channel helps to detoxify Al. Part of the old root channels is also filled with topsoil with a higher organic matter content. Various organic acids can complex Al (Hue *et al.*, 1986). Alternatively, mechanical impedance might prevent root penetration into the subsoil and the loose soil might form an easy pathway into the soil, which is probably air-filled most of the time. The symptoms seen on the maize root, however, suggest Al toxicity as a primary obstacle to root development in the subsoil rather than mechanical impedance.

The crop roots follow the old tree root channel and only occasionally form branch roots into the subsoil. As the contact between roots and surrounding soil is incomplete, the effectiveness of uptake of water and nutrients by these roots is undoubtedly reduced (De Willigen and Van Noordwijk, 1987).

Water infiltration

Old root channels are important for water infiltration under a high rainfall regime. Ponding of water at the surface may be necessary to get macropores water-filled, but such conditions are common during showers. The stain patterns observed deviate from the homogeneously colored shape predicted by soil physical models (Heinen and Raats, 1990). In some cases, such as Figure 3A, a relatively simple adjustment could be made by assuming that a certain fraction of the soil volume (*e.g.* 50%) does not participate in the flow after heavy rainfall. Such a description of the soil as a two-phase system would, however, not be adequate to deal with the major old tree root channels, as their shape and branching pattern definitely plays a role (Fig. 3B and C). It seems that old tree stumps can act as a funnel,

especially if the soil next to the tree stump was compacted by the gradual expansion of the tree trunk. Through the tree roots, radiating into the subsoil, water can be spread relatively homogeneously over the subsoil.

Occurrence of old root channels

Both aspects of old tree root channels can lead to important deviations from the assumptions used for the simple leaching model used so far. Quantification of the geometry of old root channels would be a first step, but this is obviously no simple task. A first question would be which trees of the forest vegetation actually are deep rooted, what the density of such trees is in the forest, what the life expectancy is of these trees, or individual roots of these trees, and how long old root channels remain open. Some information of this kind was given by Jones (1955) for a forest reserve near Benin city in Nigeria (about 2000 mm of rainfall, on uniform soil profiles on clayey sands, pH 4.7–6.1). 'Although it may be true that the root systems of trees in tropical rain forest are in general shallow, the distribution of roots as seen in the soil pits was very similar to that which may be seen in similarly well-drained English soils'. 'The main skeletal roots, 2–10 cm in diameter, are usually within 40 cm of the surface, but roots of 4 cm diameter have been recorded as deep as 100 cm and fibrous roots extend in small numbers beyond the bottom of almost every soil pit. Channels formed where roots have decayed were observed in several pits at depths of 80–130 cm; they become filled with loose earth of slightly duller colour than the surrounding soil and give the impression of merging with it very quickly; they were frequent and conspicuous only in two places where the forest had suffered from relatively recent windbreaks.'

Consequences for N management

With regard to the relevance of the old tree root channels for N management a key question is whether or not the water flushing through the channels carries nitrogen. If nitrogen is present in the micropores of the bulk soil, water flowing through macropores may bypass the nitrogen, as

described by Elliott and Coleman (1988) for biopores in no-till agricultural plots. For fertilizer N, recently applied at the soil surface, however, leaching can be deeper by this preferential flow than otherwise would occur. If N penetrates through the old tree root channels, crop roots still have a chance of intercepting it. In balance, we may expect that the presence of old tree root channels can be an advantage for N management. To make better use of them, however, the cropping pattern may need reconsideration. Old tree root channels may be expected to occur in a radial pattern around old tree stumps. Cropping practices where specific crops are planted next to old tree stumps, as commonly used for Yam in S. Nigeria, may increase the chance for the crop roots to find the channels leading into the subsoil. Linear crop arrangements, a common sign of modernization, might be disadvantageous in this respect. In managing the soil the connection between deep root channels and the soil surface should be maintained. Soil tillage might be harmful in this respect. In the long run, linear zones of old tree root channels can be created if hedgerow intercropping (or alley cropping, Kang *et al.*, 1989) systems are modified to include a rotation of hedgerow positions.

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